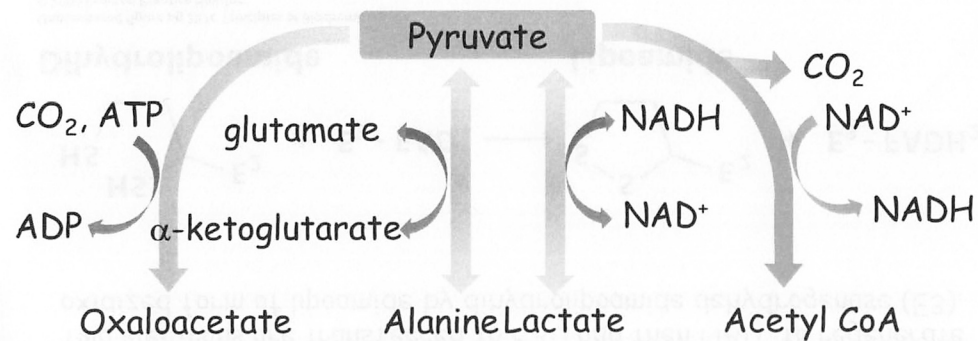
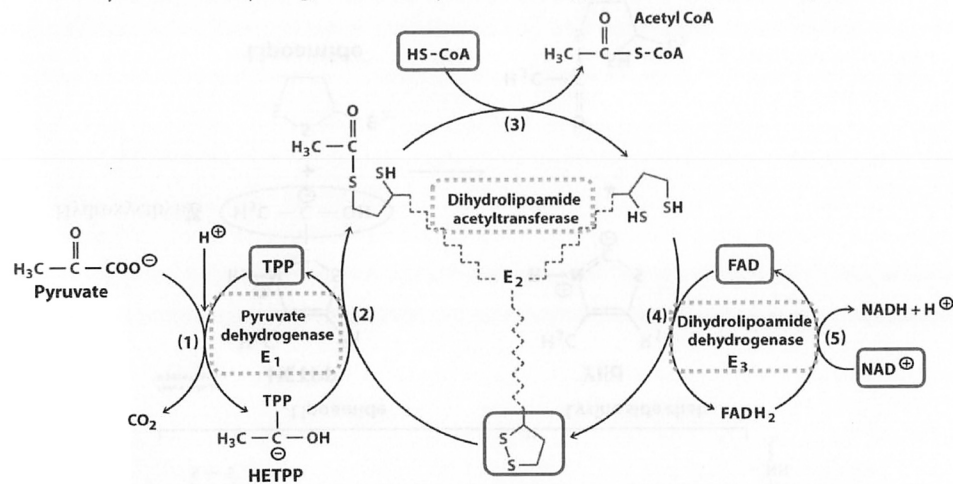


Fate of pyruvate in mammals

1. Anaerobic reduction to lactate
2. Carboxylation to oxaloacetate
3. Transamination to alanine
4. Oxidative decarboxylation to acetyl CoA



Pyruvate dehydrogenase complex is a large multienzyme complex

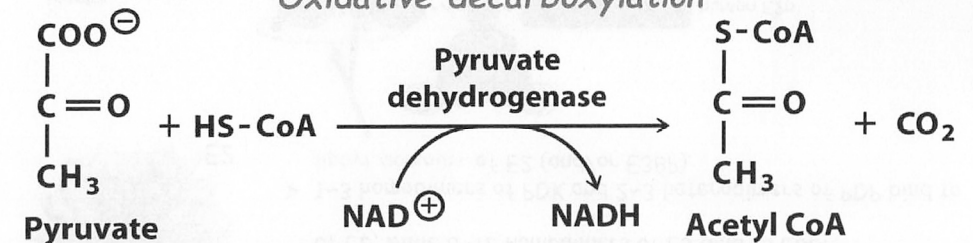


- PDC is composed of three catalytic (E1, E2, E3), one binding (E3 binding protein, E3BP) and two regulatory (PDH kinase, PDH phosphatase) components.
- PDC catalyzes 5 sequential reactions.
- PDC requires 5 coenzymes, three of which (TPP, lipoamide, FAD) are bound tightly to respective subunit.

Pyruvate is metabolized irreversibly to Acetyl CoA by pyruvate dehydrogenase complex in mitochondrial matrix

1. Entry into TCA cycle for ATP production (gatekeeper of TCA cycle)
2. Conversion of glucose to FA via production of cytosolic acetyl CoA

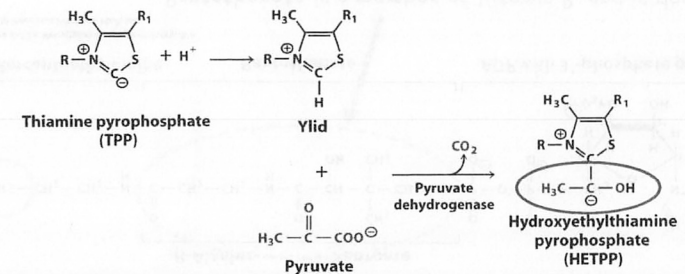
Oxidative decarboxylation



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$$\Delta G^0 = -33.4 \text{ kJ/mol}$$

TPP, a derivative of vitamin B1, is required for decarboxylation reaction of PDH (E1)



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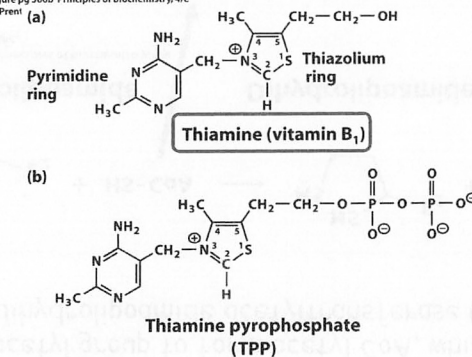
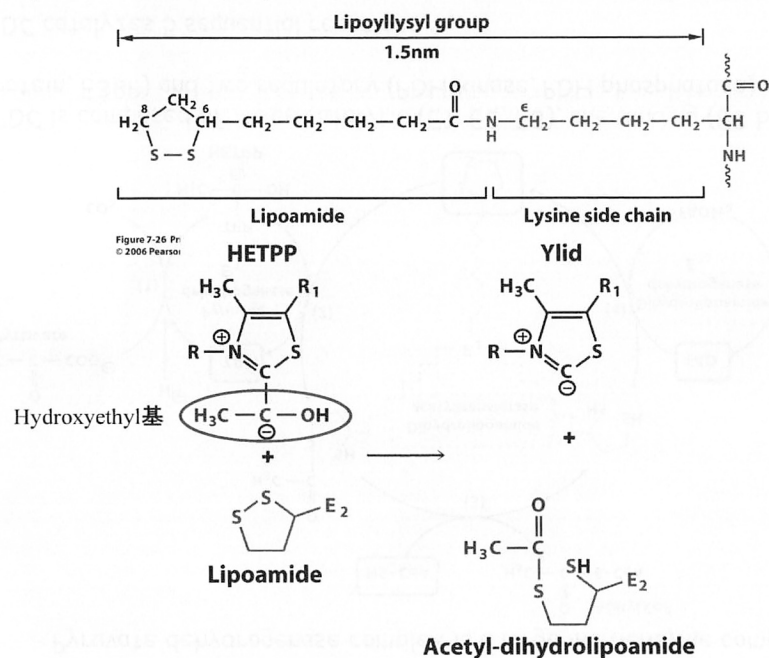
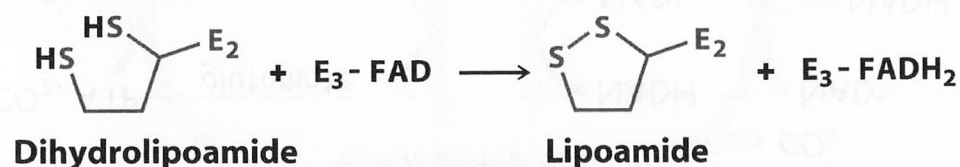


Figure 7-14 Principles of Biochemistry, 4/e
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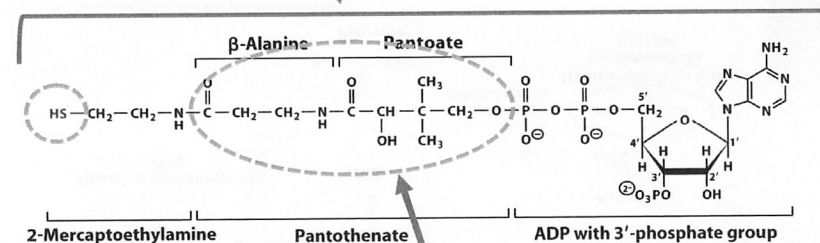
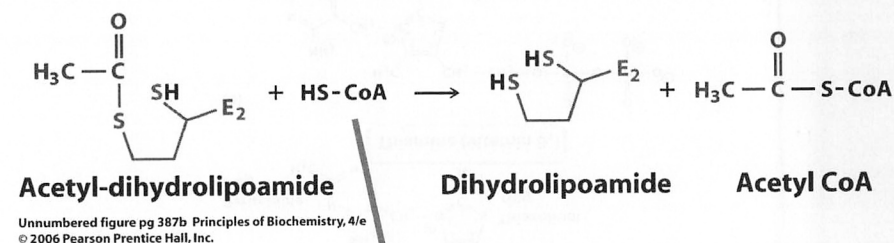
Lipoamide linked to side chain of lysine residue of dihydrolipoamide acetyltransferase (E2) serves as acceptor of acetyl group from HETPP



Two electrons are transferred to FAD and then NAD^+ to regenerate oxidized form of lipoamide by dihydrolipoamide dehydrogenase (E3).

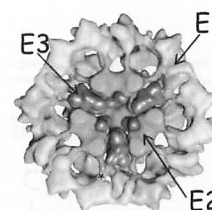


CoA receives acetyl group to form acetyl CoA, which is catalyzed by dihydrolipoamide acetyltransferase (E2)

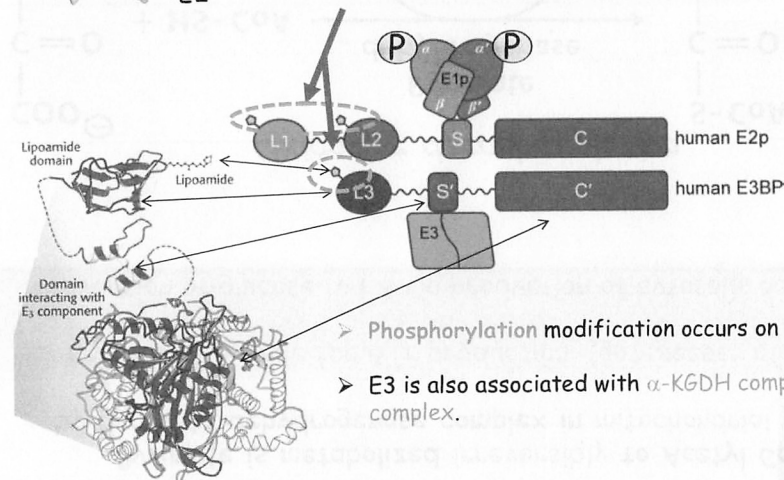


Pantothenate is a member of Vitamin B, and is derived from CoA and phosphopantothenate derivative in food.

Mammalian PDC (10, 000kDa) contains E1($\alpha_2\beta_2$), E2, E3 (α_2) and E3BP



- E2 and E3BP constitutes a central core of PDC.
- 20~30 heterotetramers of E1 bind to subunit-binding domain of E2, while 6~12 homodimers of E3 bind to E3BP.
- 1~3 homodimers of PDK and 2~3 heterodimers of PDP bind to lipoyl domains of E2 (and/or E3BP).

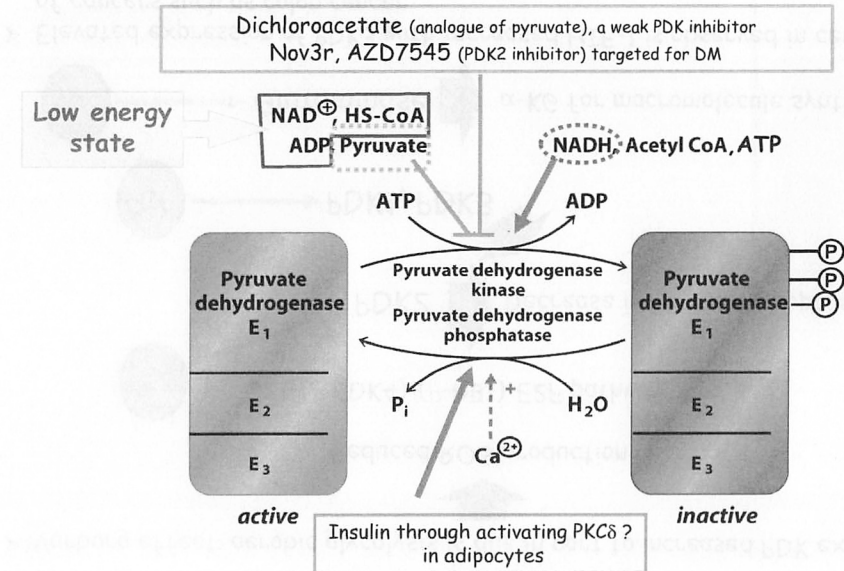


PDC activity is regulated in two ways

1. Covalent modification

2. Feedback inhibition by end products, NADH and acetyl CoA, whereas feed-forward activation by NAD⁺ and CoA

Covalent modification of PDH (E1 α subunit) determines its enzymatic activity



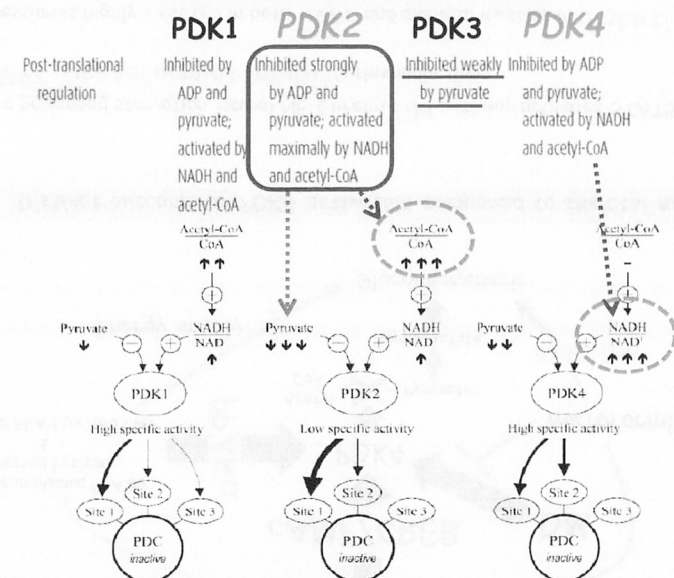
Characters of PDHK isoforms in fed-normal mice

Tissue distribution	PDK1	PDK2	PDK3	PDK4	Relative protein levels
Brain	++	+++	+++	+	PDK2 » 1 » 4
Heart	+++++	++++		++++	
Sk. muscle	+	+++		+	
Pancreas	+	+++			
Liver	++	+++			PDK1 ≈ 2 PDK2 ≈ 3
Kidney	++	+++	++++	+	
Testis	++	++	+++		
Spleen		+			
Lung	+	+	++		
Site1	63	125	36	70	BAT highly expresses PDK4 even compared to heart, but PDK2 contributes to PDC inactivation to a similar extents
Site2	20	32	74	41	
Site3	24	0	0	0	

➤ PDK2, 1 and 3 mainly contributes to inhibit PDC activity in heart, liver and kidney, respectively, in mice under fed conditions.

- PDH has three phosphorylation site [Ser293 (site 1), Ser 300 (site 2), Ser232 (site 3)].
- Phosphorylation of only one site is enough for PDH inactivation.
- Site 1 phosphorylation is the most rapid, whereas site 3 phosphorylation the least rapid, indicating importance of site 1 for acute regulation by metabolic demands.

Characters of PDHK isoforms



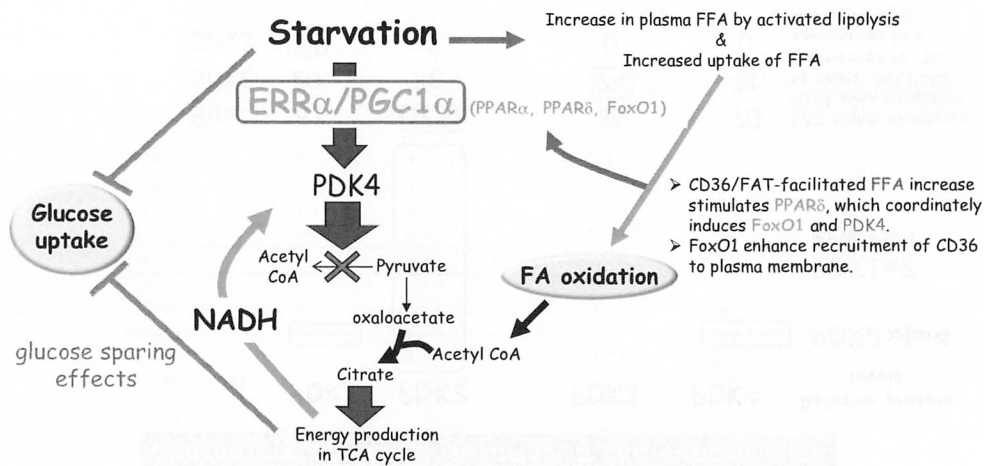
- PDK2 is most sensitive to metabolite changes, serving as a potent short-term inhibitor by phosphorylating site 1 of PDC.
- Expression of *PDK4 is induced and most sensitive to fasting and diabetes*.

Regulation of PDK skeletal muscle

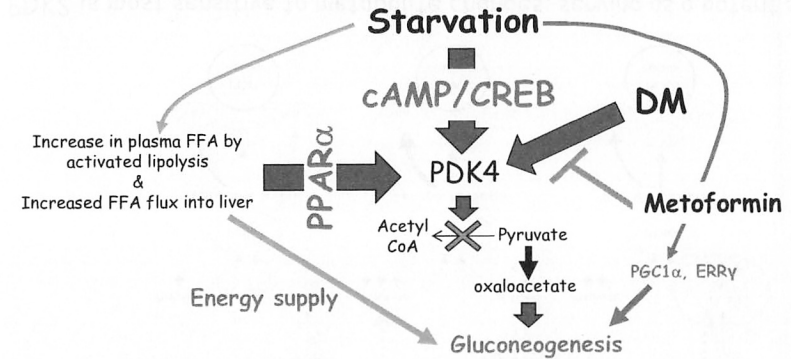
~80% of insulin-mediated glucose disposal occurs in skeletal muscle in human

In starvation, **glucose sparing effects** (limited usage of glucose by increased FA oxidation) is achieved by activation of PDK4

➤ PDK4 induction evokes DIO-induced glucose tolerance and IR



Regulation of PDK in Liver



Distinct outcome of PDK4 activation compared to skeletal muscles!!

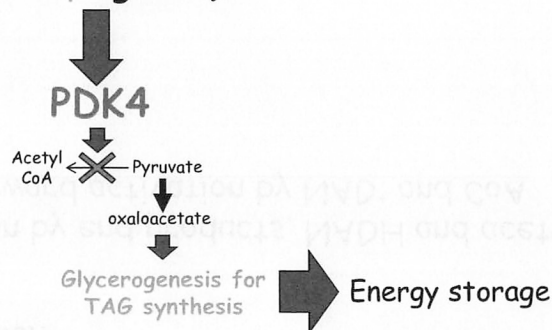
➤ In response to prolonged starvation, neural NE-Ghrelin-GH pathway activates STAT5, resulting in PDK4 induction. **AMPK (metoformin)** inhibits this induction.

➤ PDK4 expression is highly elevated in both **liver** and skeletal muscles in **diabetic conditions**.

Regulation of PDK in white adipose tissue

In WAT

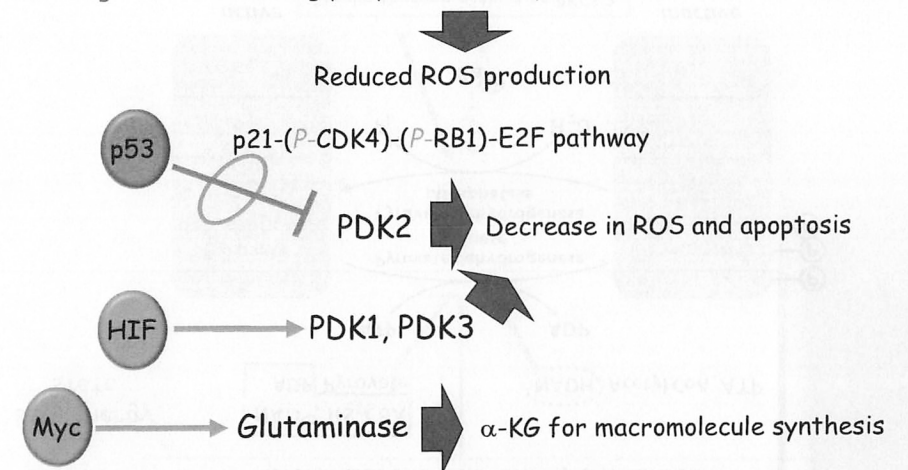
Thiazolidinediones
(PPAR_γ agonist)



- Pyruvate is utilized for glyceroneogenesis to produce DHAP and finally glycerol 3-phosphate in WAT.
- Thiazolidinediones increase PDK4 expression in WAT, but not in liver and skeletal muscles.

Regulation of PDK in Cancer

➤ Warburg effect: aerobic glycolysis is due in part to increased PDK expression



- Elevated expression of PDK3 with increased HIF-1 is observed in certain types of cancers such as colon cancer.
- Myc supplies TCA intermediates via increased glutaminolysis to support enhanced demands for macromolecule synthesis under HIF-1-mediated PDK expression

Characters of PDP isoforms

	PDP1	PDP2
Tissue distribution		
Brain	++++	++++
Heart	+++++	+++++
Sk. muscle	++++	
Pancreas		
Liver		+++++
Kidney		+++++
Testis	+++	
Spleen		+
Lung	+	

- PDP consists of two subunits, a catalytic and regulatory subunit, although regulatory subunit of PDP2 is not determined.
- Dephosphorylation of three phosphorylation sites occurs randomly by PDP.
- PDP1, but not PDP2, is stimulated by Ca^{2+} .

Highly reductive ($NADH/NAD^+$) and Acetylated (Acetyl CoA/CoA) lipoyl domain of E2 allosterically activates binding of PDHK to lipoyl domain of E2.

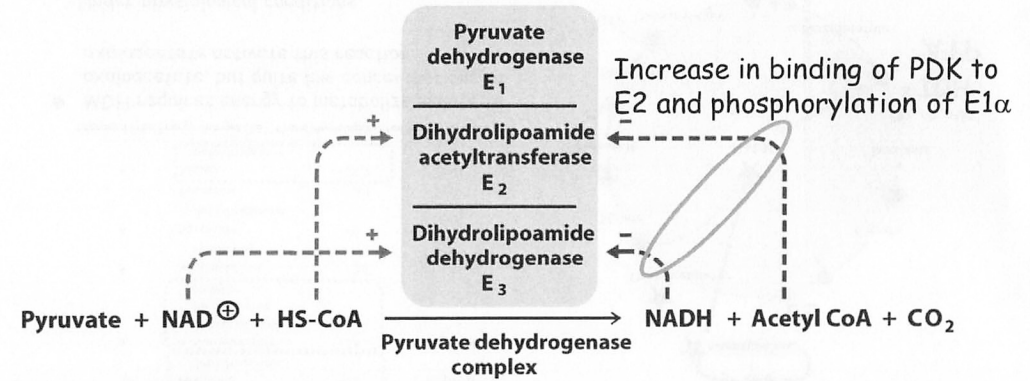
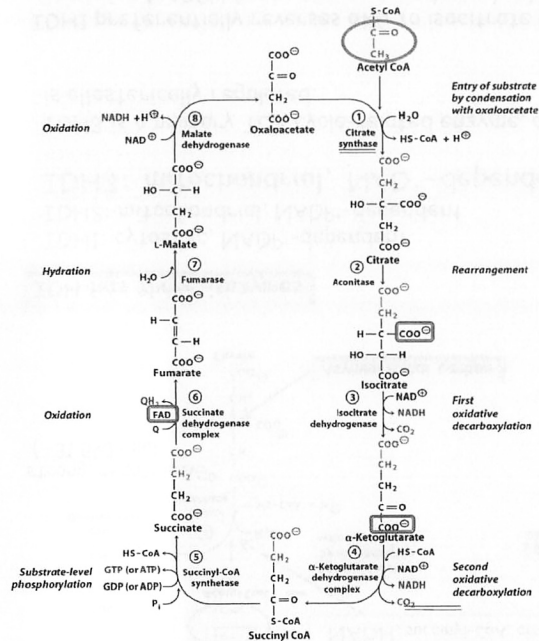


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TCA cycle oxidizes acetyl CoA

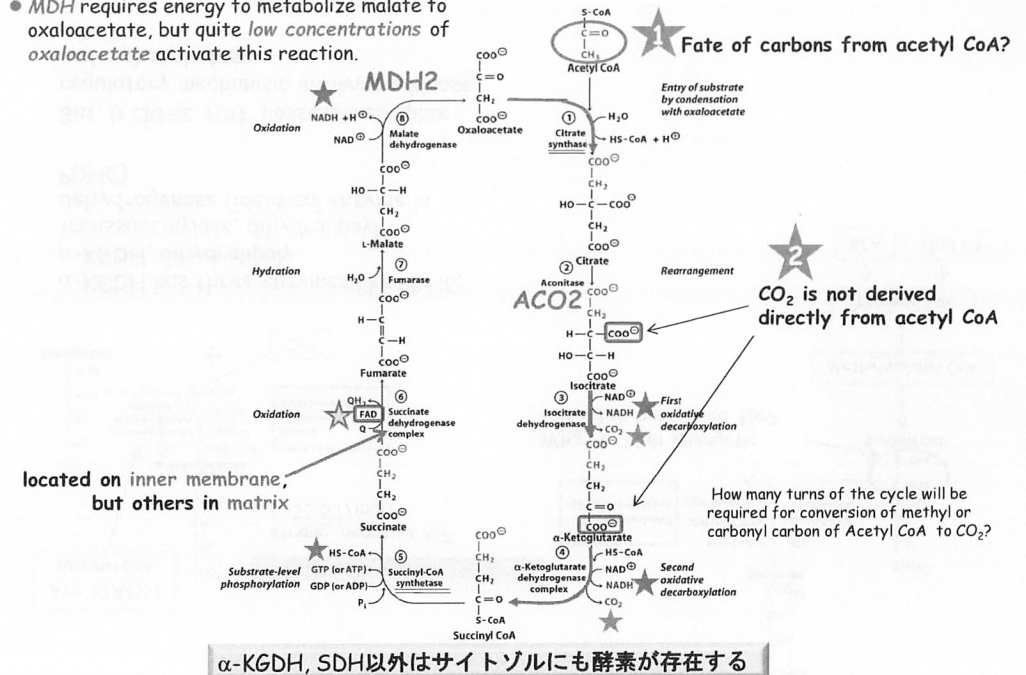


- TCA cycle neither consumes molecular oxygen nor produces meaningful amounts of ATP.
- TCA cycle removes electrons from acetyl CoA and transfers to $NADH$ and $FADH_2$, electron carriers, and subsequently to mitochondrial ETC.
- Acetyl CoA is provided from
 1. glucose-derived pyruvate
 2. lactate-derived pyruvate
 3. fatty acid oxidation
 4. amino acid (e.g. Lys, Leu, Ile)
 5. ketone body and acetate
- Oxaloacetate (OAA) serves as a catalyst in TCA cycle, because OAA is regenerated.

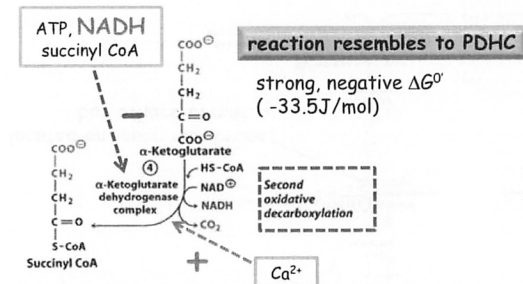
Figure 13-3 Principles of Biochemistry, 4/e
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TCA cycle oxidizes acetyl CoA

- MDH requires energy to metabolize malate to oxaloacetate, but quite low concentrations of oxaloacetate activate this reaction.



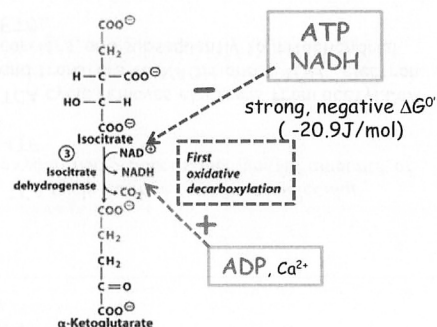
α-KGDH, SDH以外はサイトソルにも酵素が存在する



IDH1: cytosolic, NADP⁺-dependent
IDH2: mitochondrial, NADP⁺-dependent
IDH3: mitochondrial, NAD⁺-dependent

IDH3 is a primary TCA cycle-related enzyme, and is allosterically regulated.

IDH1 preferentially reverses α KG to isocitrate by consuming NADPH (reductive carboxylation), while IDH2 mainly serves as an oxidative TCA cycle enzyme.

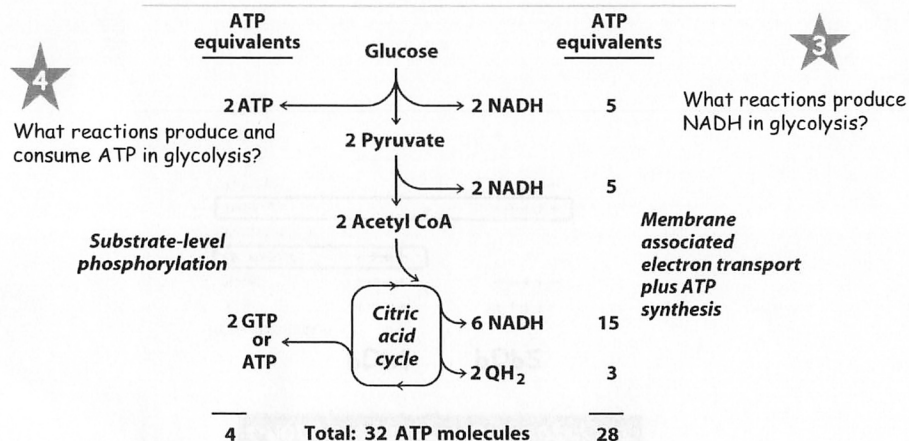


α -KGDH has three enzymes like PDHC;
 α -KGDH, dihydrolipoyl
transsuccinylase, dihydrolipoyl
dehydrogenase (identical enzyme in
PDHC)

But, it **dose not** possess a complex regulatory mechanism involving a kinase and a phosphatase.



Reaction	Energy-yielding product	ATP equivalents
Isocitrate dehydrogenase	NADH	2.5
α -Ketoglutarate dehydrogenase complex	NADH	2.5
Succinyl-CoA synthetase	GTP or ATP	1.0
Succinate dehydrogenase complex	QH ₂	1.5
Malate dehydrogenase	NADH	<u>2.5</u>
Total		10.0



Regulation of TCA cycle is regulated by cellular energy requirement

Reaction	Enzyme	ΔG° ($\text{kJ} \cdot \text{mol}^{-1}$)	ΔG ($\text{kJ} \cdot \text{mol}^{-1}$)
1	Citrate synthase	-5	Negative
2	Aconitase	-31.5	~ 0
3	Isocitrate dehydrogenase	-21	Negative
4	α -Ketoglutarate dehydrogenase multienzyme complex	-33	Negative
5	Succinyl-CoA synthetase	-2.1	~ 0
6	Succinate dehydrogenase	+6	~ 0
7	Fumarate dehydrogenase	-3.4	~ 0
8	Malate dehydrogenase	+29.7	~ 0

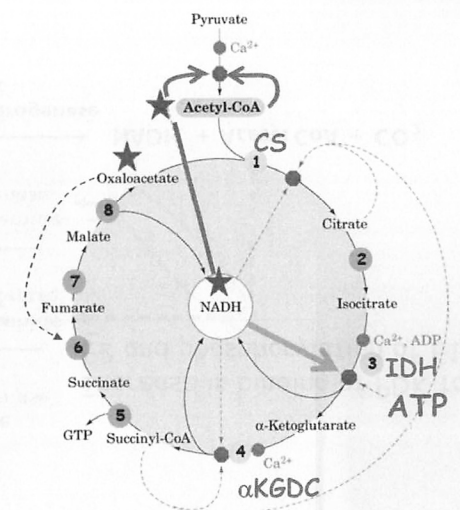
Standard Free Energy Changes (ΔG° , kJ) and Physiological Free Energy Changes (ΔG)

- MDH requires energy to metabolize malate to oxaloacetate, but quite *low concentrations* of *oxaloacetate* activate this reaction.

Under physiological conditions,
1. Citrate synthase is regulated by its substrate concentration (acetyl CoA and oxaloacetate).

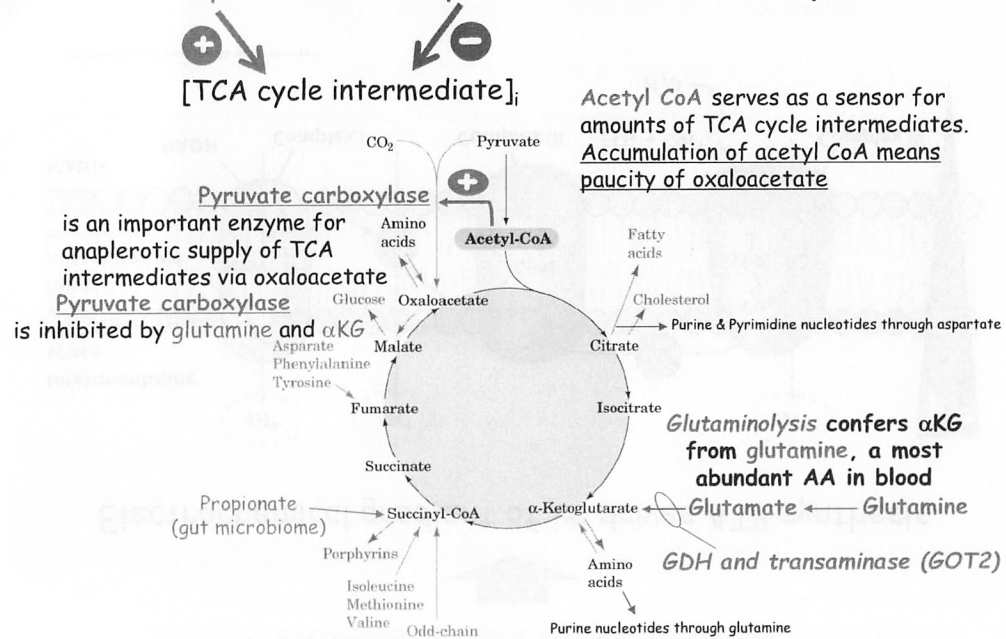
2. α -KGDC is **not** regulated by covalent modification like PDC

- In *skeletal muscles*, increase in Ca^{2+} also plays crucial roles by activating PDH, IDH and αKGDH



Most important allosteric effectors are **NADH**, **acetyl CoA** and **Oxaloacetate**

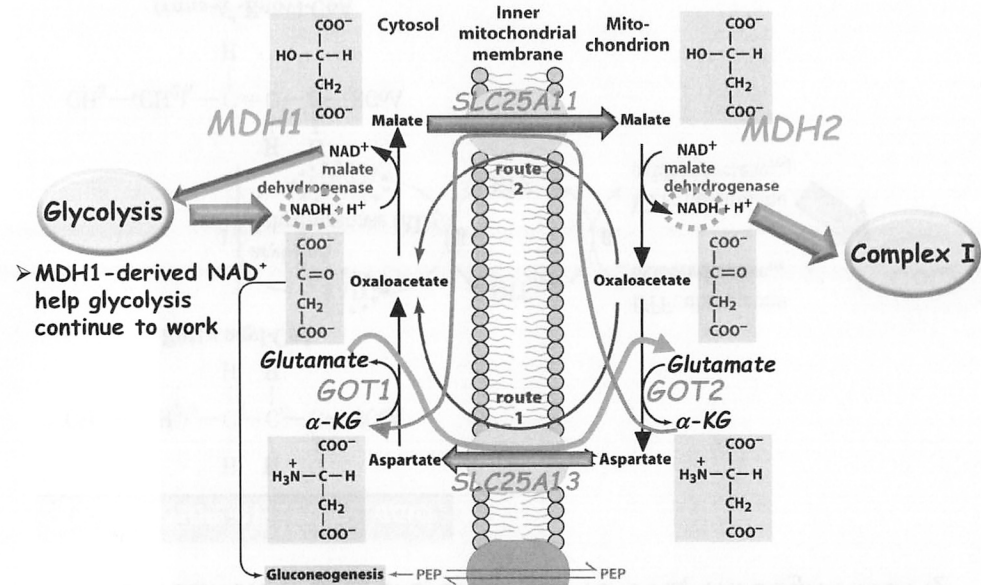
Anaplerotic and Cataplerotic reaction of TCA cycle



- Pyruvate carboxylase and Glutaminolysis are two major pathway replenish TCA intermediate (anaplerotic reactions)

MDH-mediated Malate-Aspartate shuttle

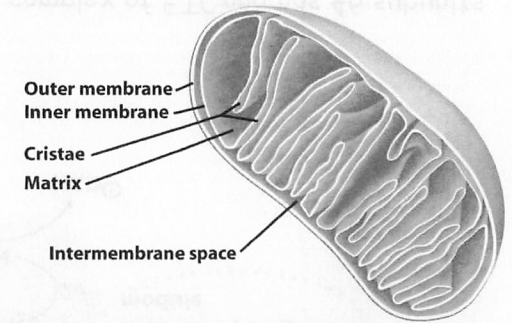
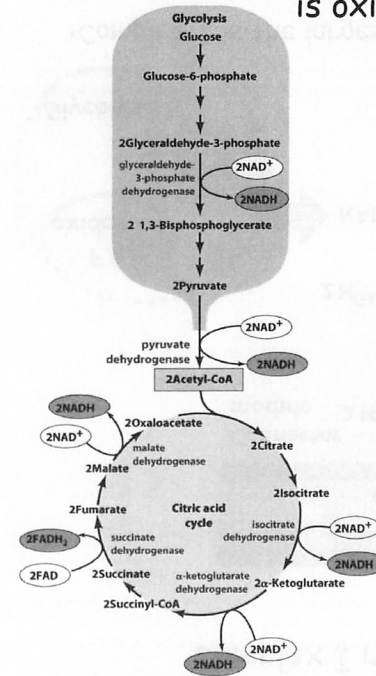
transfers NADH from cytosol to mitochondria



This shuttle is also important for oxaloacetate transfer for gluconeogenesis

- MDH1 also play crucial roles in conjunction with malic enzyme in de novo FA synthesis

Glycolysis produces 2NADH in cytosol, but how these NADH is oxidized in mitochondria?

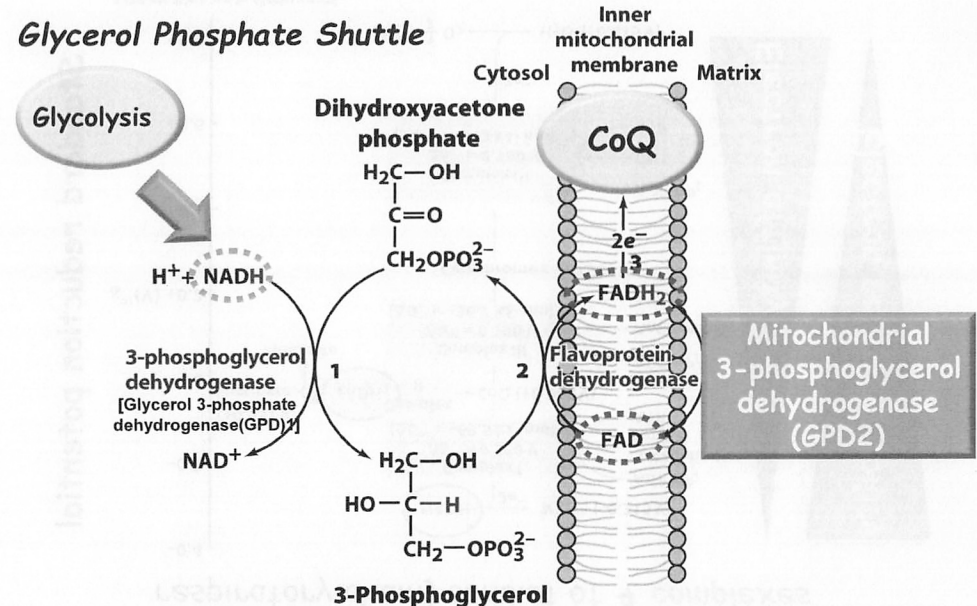


➤ Porin (VDAC), a protein embedded in Mit outer membrane, can diffuse ions and molecules with less than 10kDa freely, resulting in equilibrium between cytosol and intermembrane space.

➤ On the other hand, inner membrane has much lowered permeability against ions and metabolites, leading to establishment of steep gradient between the membrane.

Mitochondrial Flavoprotein dehydrogenase feeds electron from cytosolic NADH to CoQ through FADH₂

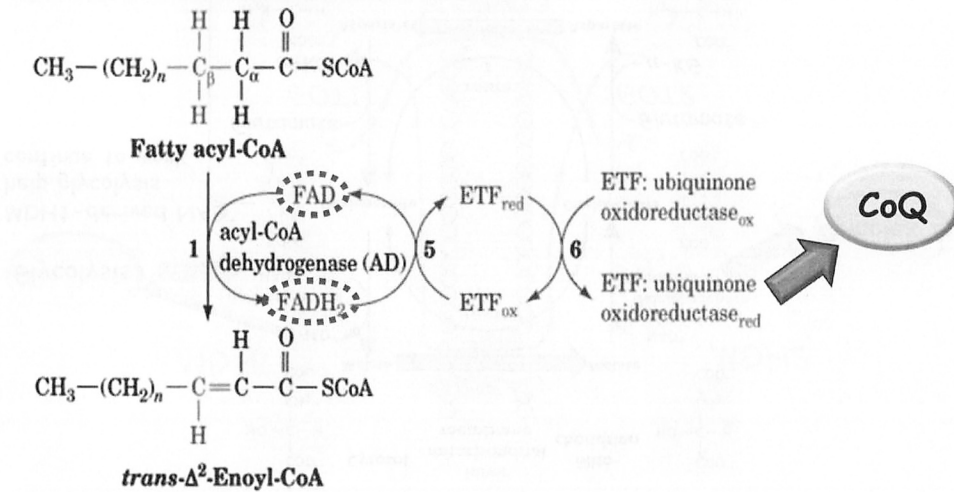
Glycerol Phosphate Shuttle



➤ The shuttle is active in BAT and involved in thermogenesis

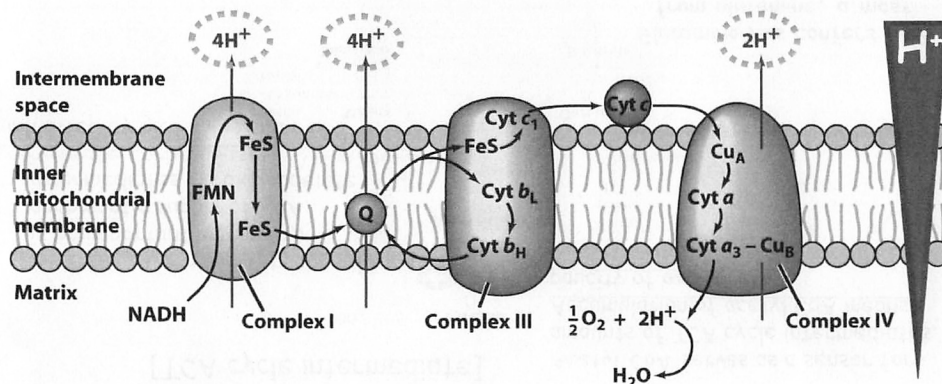
Mitochondrial Flavoprotein dehydrogenase responsible for FA oxidation transfers electron to **CoQ** through **FADH₂**

β oxidation of Fatty acid



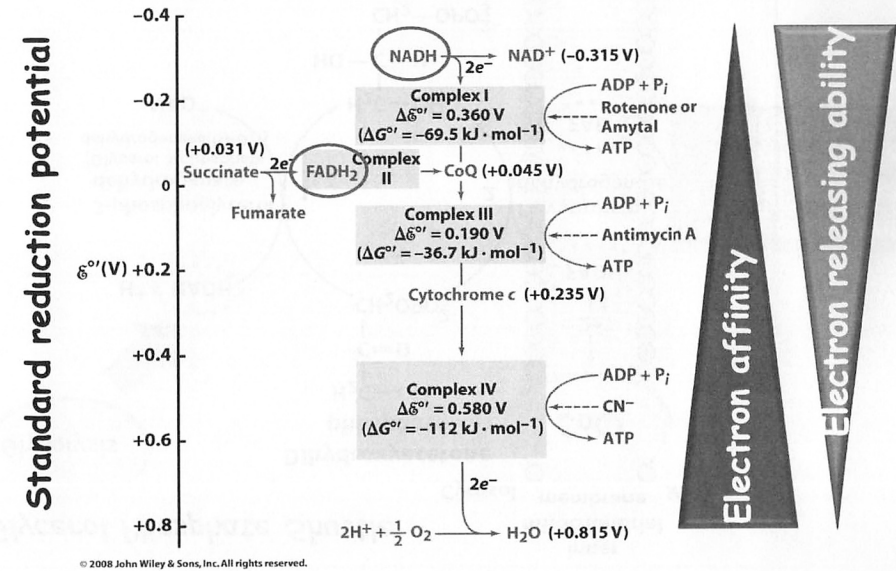
Transfer of electrons through ETC pumps **H⁺** from matrix to intermembrane space

Electrochemical gradient of **H⁺** drives ATP synthesis



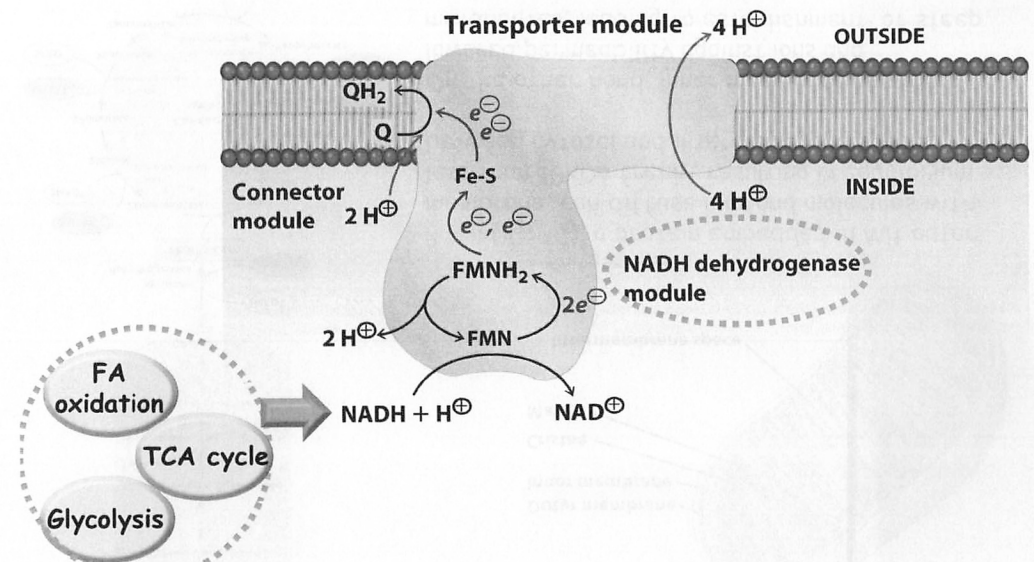
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The electron transport chain (ETC, known as a mitochondrial respiratory chain) consist of **4 complexes**



Complex I: NADH-ubiquinone oxidoreductase, Complex II: succinate-ubiquinone oxidoreductase
Complex III: ubiquinol-cytochrome c reductase, Complex IV: cytochrome c oxidase

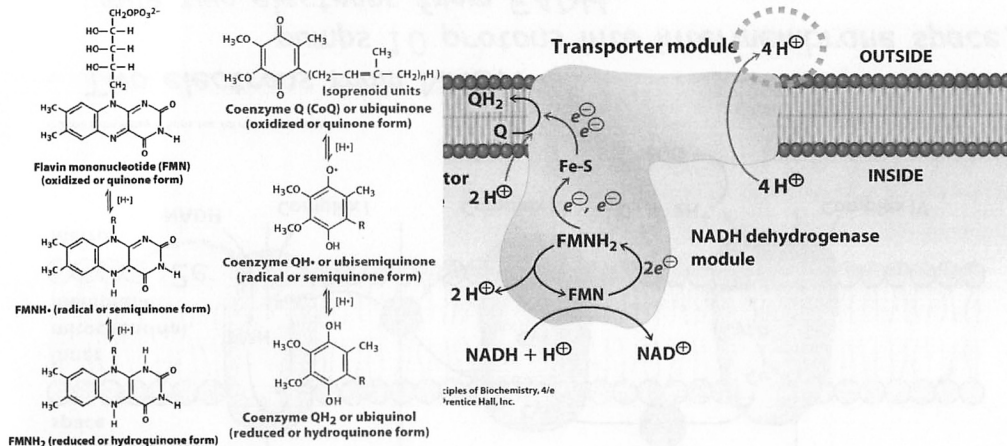
Complex I receives **2 electrons** from **NADH**



•Complex I is the largest protein complex of ETC and has 46 subunits.

•Complex I has one **FMN** and several **FeS** clusters (8-9).

Complex I pumps 4 protons into IMS



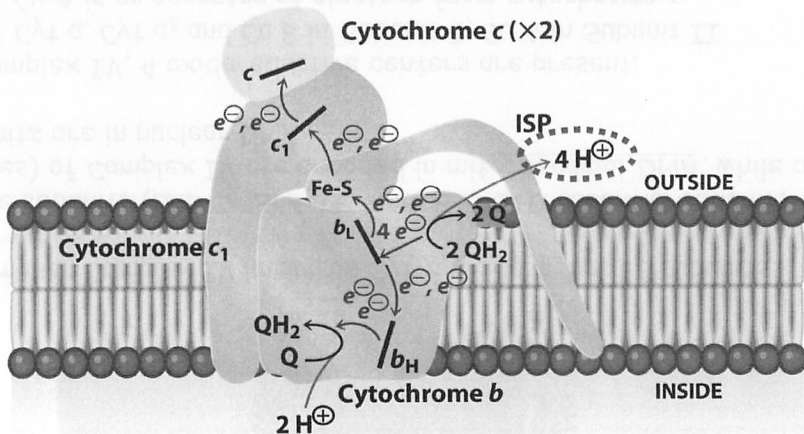
• FMN and CoQ can serve as either one or two electrons donor or acceptor.

• FeS cluster can receive and release only one electron at a time.

• During 2 electrons transfer,

4H⁺ are pumped out into IMS

Complex III transfers electron from CoQ to cytochrome c, a mobile electron carrier, between Complex III and IV

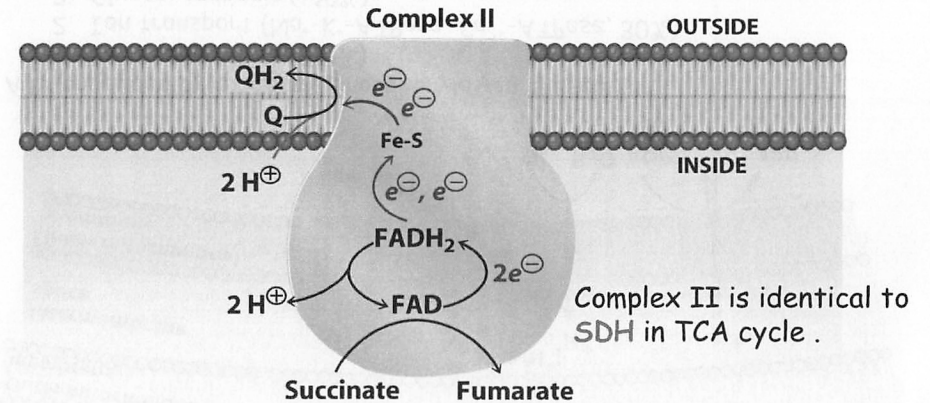


• Complex III (known as cytochrome bc₁, 11 subunits) has one unique FeS clusters known as Rieske iron-sulfur protein (ISP), one cytochrome c₁, and 2 b-type cytochromes (b_H and b_L).

• During 2 electrons transfer,

4H⁺ are pumped out into IMS by Q cycle

Complex II transfers 2 electrons from FADH₂ to CoQ



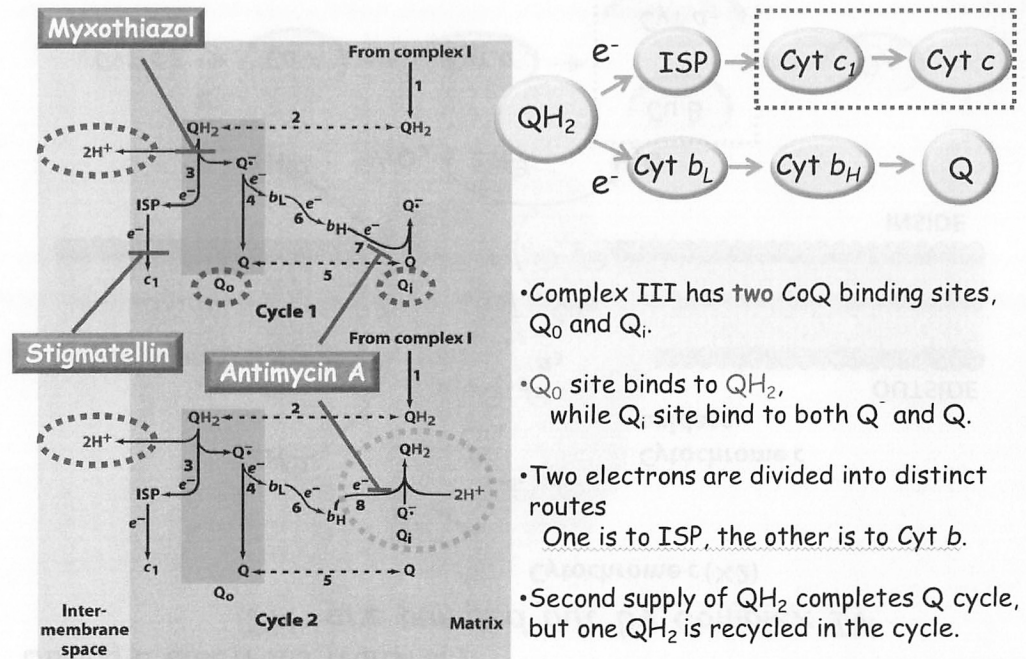
• Complex II has a covalently bound-FAD, 3FeS clusters and one cytochrome (b560).

• During the electron transfer, **H⁺ is not pumped out into IMS**



Not involved in H⁺ gradient and ATP synthesis

Q cycle supports electron transfer and proton pumping in Complex III



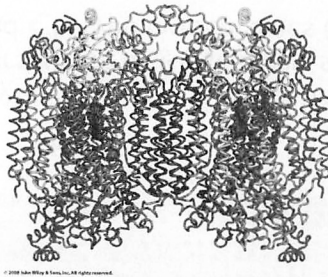
• Complex III has two CoQ binding sites, Q₀ and Q_i.

• Q₀ site binds to QH₂, while Q_i site bind to both Q⁻ and Q.

• Two electrons are divided into distinct routes
One is to ISP, the other is to Cyt b.

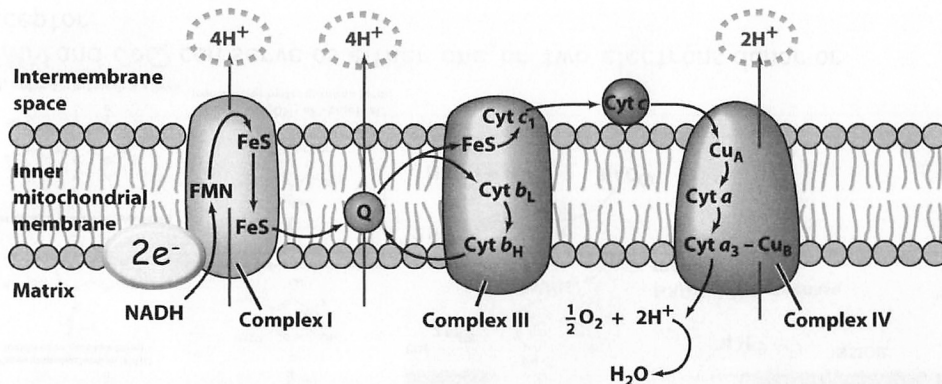
• Second supply of QH₂ completes Q cycle, but one QH₂ is recycled in the cycle.

Complex IV transfers electrons from cytochrome c to Oxygen, the terminal electron acceptor



- Mammalian Complex IV known as Cyt c oxidase has 13 subunits.
- Three subunits (Sub I, Sub II, and Sub III: essential subunits in all species) of Complex IV are encoded in mitochondrial DNA, while other subunits are in nuclear DNA.
- In Complex IV, 4 oxidoreductive centers are present;
 - ✓ Cyt a, Cyt a₃ and Cu B in Subunit I, Cu A in Subunit II
 - ✓ Cu A is an acceptor of electron from cytochrome c
 - ✓ Cyt a₃ and Cu B form a binuclear center involved in electron transfer to oxygen

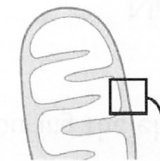
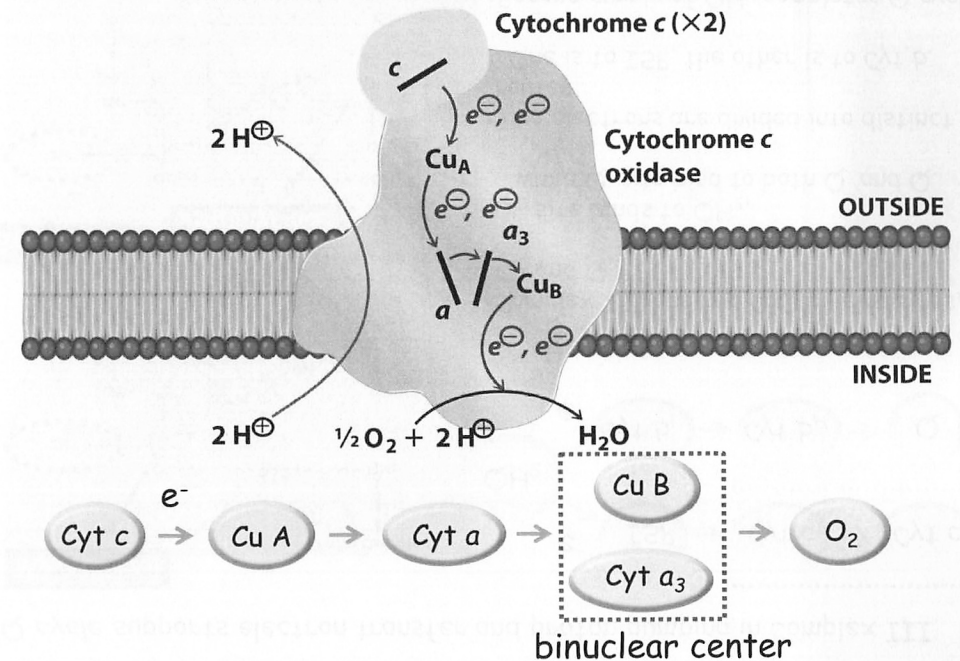
**Proton transfer occurs at three sites;
Complex I, CoQ-Complex III, and Complex IV**



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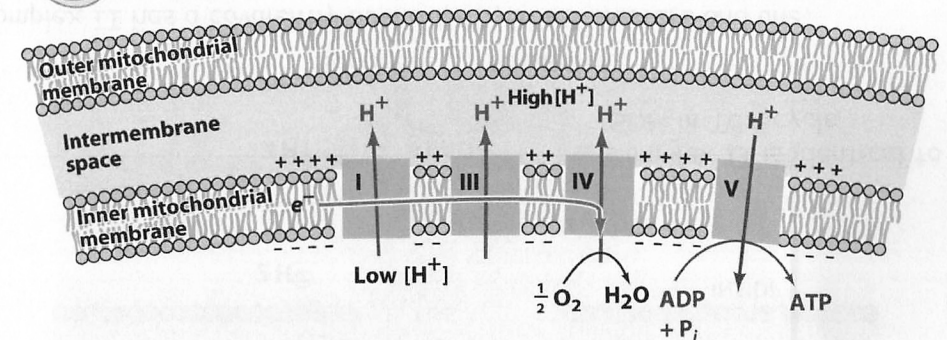
**Two electrons from NADH
pumps 10 protons into intermembrane space,
while two electrons from FADH₂
pumps 6 protons into IMS**

During 2 electrons transfer,
2H⁺ are pumped out by Complex IV



**Oxidative Phosphorylation
by ATP synthase (Complex V)**

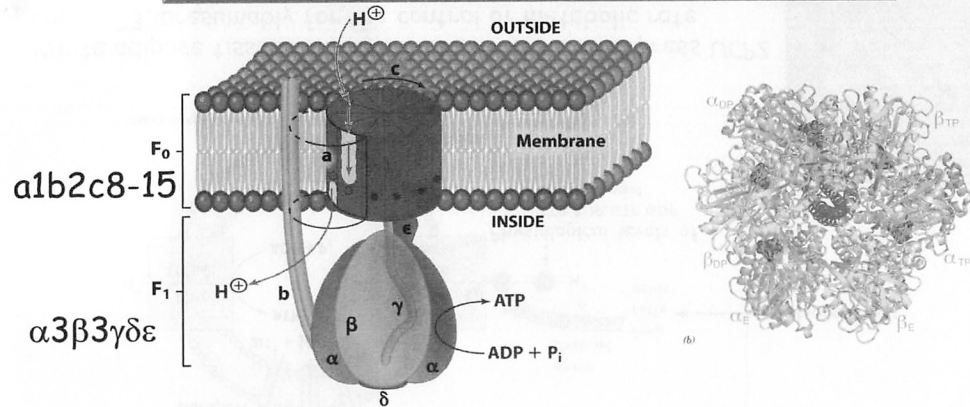
Electrochemical gradient of H⁺ drives ATP synthesis



ATP produced by oxidative phosphorylation is used for

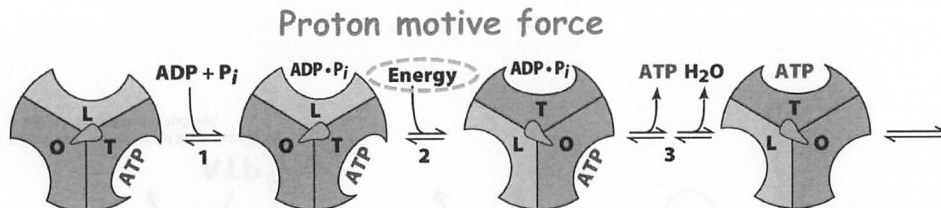
1. Protein synthesis (30%)
2. Ion transport (Na⁺-K⁺-ATPase, Ca²⁺-ATPase, 30%)
3. Gluconeogenesis (~10%)
4. Heat, etc

ATP synthase consists of two domains, F_1 and F_0



- Numbers of c-subunit vary depending on species. Mitochondria have 8 c-subunits, while E.coli has 12 c-subunits.
- Human and rodent mitochondria have additional 5 subunits to form F_0 complex, which are thought to have regulatory functions.
- α and β subunit shows high similarity in AA sequences and its structure.
- α and β subunit binds with either ADP and ATP, but only β subunit shows ATPase activity.
- F_1 portion is protruding into cytosol in bacteria, while its protrusion faces matrix side in mitochondria.

A model of ATP synthesis proposed by Paul Boyer



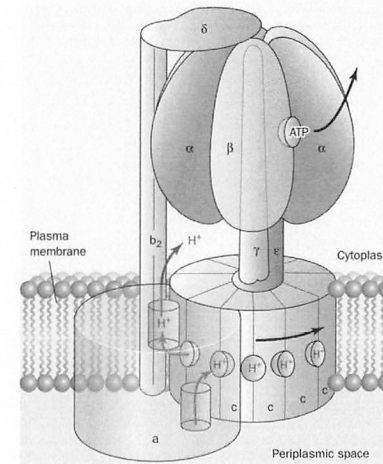
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- ab protomer shows 3 distinct states (O, open; L, loose; T, tight), but only b subunit has catalytic activity site (ATPase activity).

- Each b subunit in ATP synthase shows different state, and works cooperatively, but not independently. Therefore, ATP synthase can produce a large amount of ATP in a short time.

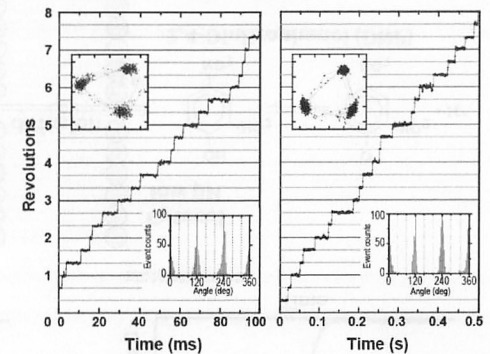
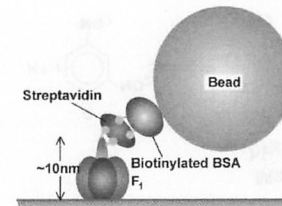
- Proton motive force causes **conformational changes** in ATP synthase, resulting in the release of tightly-bound ATP from T form, binding of ADP and P_i in a L form, and converting L form to T form.

ATP synthase is an energy-generating rotary motor engine



- F_0 , consisting of a/c subunits, serves as a proton channel, while F_1 acts as a plug for the channel (g subunit) and generates ATP (b subunit).
- c rings and g(ϵ) stalk are rotor, whereas other molecules are stator.
- c rings and g(ϵ) stalk rotate when H^+ enters into F_0 . Interaction between Asp61 in c subunit and Arg210 in a subunit is involved in the rotation of c subunit.
- ATP synthase generates 3 mol ATP when c rings with g(ϵ) stalk rotate 360-degree. For example, 1 mol ATP production requires 3 protons when c rings consist of 9 c subunits.

ATP synthase rotates at high speed



- ATP synthase is assumed to rotate 24,000 times/min.
- Proton gradient is resolved **exclusively** by ATP generation by ATP synthase, but other ways also are involved in the resolution (e.g. transport of P_i into matrix). Therefore, circa 4 protons are required to generate ATP.

NADH pumps 10 protons,

resulting in about 2.5 ATP production

while $FADH_2$ pumps 6 protons,

resulting in circa 1.5 ATP production

Electrochemical gradient transport of ions for ATP production

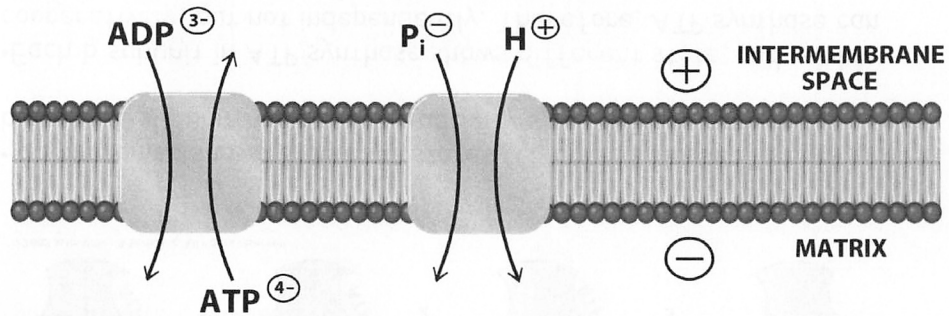


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Uncoupler disrupts connection of ATP synthesis and substrate oxidation

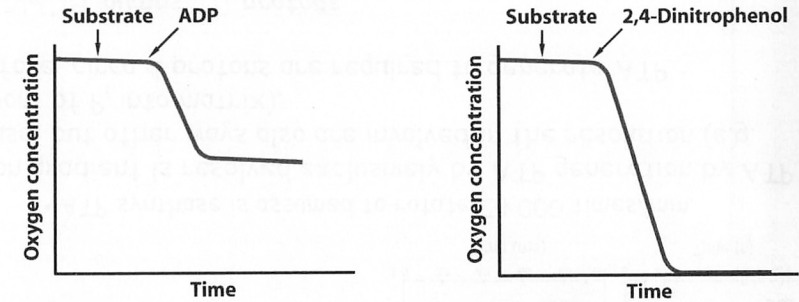
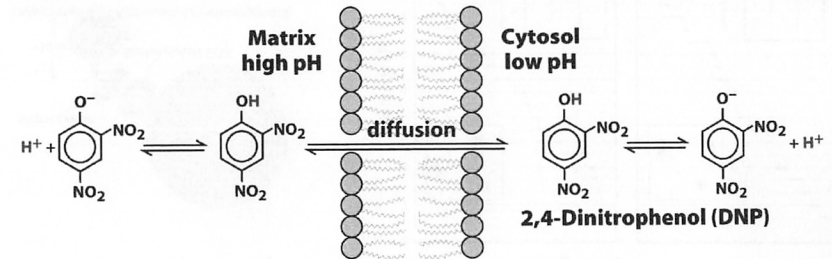


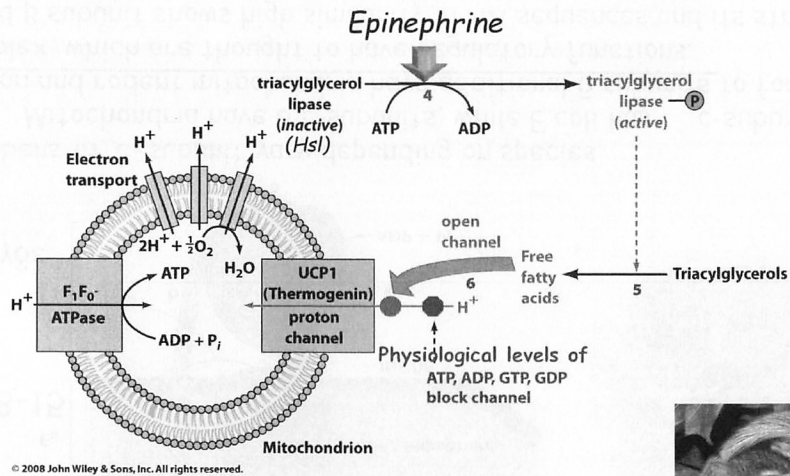
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Brown adipose tissues express physiological uncoupler protein (UCP1) to produce heat



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White adipose tissues and skeletal muscles also express UCP2 and UCP3, presumably for the control of metabolic rate